



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/608694

Date:

18/11/2022(17:36:03)

Subject: Review of LMI "BOILER HEADER LIGAMENT CRACKING – HYDRAULIC LEAK TESTING DURING BREAKDOWN REPAIRS"

OPERATION & MAINTENANCE

MECHANICAL MAINTENANCE

(PRESSURE PARTS)

REF.: SMPP/O&M/MM/PP/

Date: 18.11.2022

Subject: Review of LMI "BOILER HEADER LIGAMENT CRACKING – HYDRAULIC LEAK TESTING DURING BREAKDOWN REPAIRS"

LMI No: LMI/OGN/OPS/MECH/005

The above approved LMI was prepared based on reference document - " **COS-ISO-00-OGN/OPS/MECH/005**" circulated by CC-OS. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

Rahul Jain - 013328 (SENIOR MANAGER ,MECH MAINT- BOILER) ,Simhadri
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On:18/11/2022(17:36:03) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/608694

May please provide approval for the revised version of subject LMI document with rev-6, as per reference document COS-ISO-00-OGN/OPS/MECH/005.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

Suresh Babu Kolla - 007980 (DY. GENERAL MANAGER ,MECH MAINT- BOILER)
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On:19/11/2022(19:56:06) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/608694

The necessary changes done i.e.

1. Temperature measurement clause added
2. Cyclic hydro caluse revised and graph included.

Revised LMI (attached as annexure) may pl. be approved

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:20/11/2022(11:06:13) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/608694

Submitted for approval pls

Submitted by:

झिमलि धर - अपर महाप्रबंधक

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On:21/11/2022(13:50:53) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/608694

Revised LMI may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
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On:22/11/2022(13:24:51) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/608694

Please review the document.

Submitted by:

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,OPERATION)
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On:23/11/2022(12:03:37) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/608694

Submitted by:

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On:23/11/2022(12:08:56) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/608694

Pl review

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

Rajasekhar Pala - 005146 (ADDL. GENERAL MANAGER ,OPN (MAIN PLANT))
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On:26/11/2022(17:03:52) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/608694

Points related to Operation are noted and will be taken care of during AOH.

Submitted by:

राजेश गणपतराव मोहदिकर - उप महाप्रबंधक

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On:30/11/2022(11:42:17) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/608694

Being followed .Forwarded for further approval

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

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On:01/12/2022(17:07:36) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/608694

May please be approved.

Submitted by:

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On:01/12/2022(19:46:12) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/608694

Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

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On:08/12/2022(18:50:24) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/608694

Approved & Submitted by:

गिरीश चन्द्र चौकसे - मुख्य महाप्रबंधक

Girish Chandra Choukse - 002907 (CHIEF GENERAL MANAGER ,HEAD OF PROJECT)
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On:08/12/2022(19:24:22) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/608694

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/005/

ISSUE NO.1; REVISION NO. 6.00; DATE: Nov'2022

**BOILER HYDRAULIC LEAK
TESTING DURING BREAKDOWN
REPAIRS**

APPROVED FOR

***Implementation*.....**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	BOILER HYDRAULIC LEAK TEST DURING BREAKDOWN REPAIRS
<u>2.a) DOCUMENT CODE</u> b) REVISION/ISSUE NO.&DATE	<u>LMI/OGN/OPS/MECH/005/</u> <u>ISSUE-1; REV.-6.00; Nov'2022</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>COS-ISO-00-OGN/OPS/MECH/005</u> <u>Rev.No.1</u>
<u>4.PREPARED BY:</u> Sr. MANAGER (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM(MM-BMD) AGM (MM-BMD) GM (MTC.) GM (OPERATION) GM (O&M)	SIGNATURE_____ SIGNATURE_____ SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE_____ HOP, SIMHADRI
<u>7.DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.
<u>9.DISTRIBUTION LIST</u>	ENCLOSED

NTPC	LMI	SIMHADRI
TITLE: BOILER HYDRAULIC LEAK TEST DURING BREAKDOWN REPAIRS REV: 6.00 DATE: Nov'2022		
REF NO. LMI/OGN/OPS/MECH/005/		PAGE : 1 OF 8
<u>1.0 INDEX</u>		
<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
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2.0	INTRODUCTION	2
3.0	SCOPE	2
4.0	REFERENCE DOCUMENT	2
5.0	SUPERSEDED DOCUMENT	2
6.0	SPECIFIC INSTRUCTIONS	2
7.0	METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS	3-4
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NTPC	LMI	SIMHADRI
TITLE: BOILER HYDRAULIC LEAK TEST DURING BREAKDOWN REPAIRS REV: 6.00 DATE: Nov'2022		
REF NO. LMI/OGN/OPS/MECH/005/		PAGE : 2 OF 8
<u>2.0 INTRODUCTION</u> During breakdown and post overhaul it is common to fill and pressurize boiler to locate a tube leak or to confirm that all damaged areas have been repaired. However, during this time, the headers are likely to be still hot and there is a risk of thermal shock and consequent crack generation and propagation. To minimize the risk of damage, it is necessary to ensure that header metal temperatures are not quenched by more than 50C but units without ligament cracks quenching can be maximum up to 100C.Cyclic hydro at water temperature less than 50C is to be done for boilers operating under flexible regime. Additional temperature monitoring is to be ensured. The objective of this LMI is to detail the operation constraint that is to be adhered to while carrying out hydraulic leak test of boilers, to prevent boiler header ligament cracking. <u>3.0 SCOPE</u> The scope of this LMI is to detail the parameters to be monitored, steps to be followed and constraints that are to be rigidly adhered to while carrying hydraulic leak test of all fossil boilers. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note COS-ISO-00-OGN/OPS/MECH/005 Rev.No.1 circulated by operation services-NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> - BOILER HEADER LIGAMENT CRACKING – HYDRAULIC LEAK TEST DURING BREAKDOWN REPAIRS REV: 4.00 DATE: MAR'2020 <u>6.0 SPECIFIC INSTUCTIONS</u> The objective of the LMI is sought to be achieved through: - Extending boiler cooling times or water filling times to avoid thermal shock caused by sudden quenching in super heater headers. - Carrying out specific actions in sequence at different stages of in-service hydraulic pressure testing of boilers		

NTPC	LMI	SIMHADRI
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<u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u> While filling a boiler for hydraulic leak testing during repairs the following norms are to be observed i) Written clearance for hydraulic test to UCB after evacuating all personnel in and near the boiler. Responsibility - HOS (PP) ii) No hydraulic leak test to be carried out unless SH header temperatures have come down to 200°C. Responsibility - HOD (O) & HOD (BMD) iii) Hydraulic Leak testing of boilers to be done with water having temperature between 20⁰-50⁰ C (Reg IBR 379). Further, Drum differential temperature limits of 50° C for filling to be maintained. Responsibility - HOD (O) & HOD (BMD) iv) For Units having Super heater headers containing internal ligament cracking, the differential temperature between feed water temperature and Super heater metal temperature should not be more than 50° C .This can be ensured by adjusting the incoming feed temperature and monitoring the feed water temperature and metal temperature of drum. Responsibility - HOD (O) & HOD (BMD) v) For Units not having Super heater headers containing internal ligament cracking, the differential temperature between feed water temperature and Super heater metal temperature should be limited preferably to 50°C but in no case be above 100 ° C during filling. Responsibility - HOD (O) & HOD (BMD) vi) All the drum vents, superheater vents are to be in open condition. All the superheater drains, divisional panel O/L to soot blowing line isolation v/v, water wall low point drains, sampling line isolation v/v's, SH & RH spray line isolation v/v's, boiler stop v/v's, MS line drains, ERV isolation v/v's & integral bypass v/v's, CBD & EBD isolation v/v, phosphate dosing line isolation v/v to be in closed condition. All the turbine side drains should be in open condition, to drain water in case of any passing of boiler stop v/v. Responsibility - HOD (O) vii) Start Boiler Fill pump and fill the boiler. Pressurize the boiler with MDBFP/ portable reciprocating pump and gradually raise the pressure. Pressure raise should not be more than 10 ksc/min up to 80% of max. working pressure (of drum) and after that it should be limited to 2 ksc/min. The drum vents & super heater vents are to be closed as and when the water comes fully from the corresponding pipes. Responsibility - HOD (O) & HOD (BMD)		

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viii) At 80% of popping pressure of superheater safety valves i.e., at 145-150 ksc hold the pressure for installation of gags. Responsibility - HOD (O) ix) Gag all the spring-loaded safety valves of drum and superheater (all 08 no.) and communicate the same to UCB in writing. Responsibility - HOS (PP) x) Raise the pressure to the value as desired by MMD, which shall be less than the maximum allowable safe working pressure and hold for 10 minutes. Responsibility - HOD (O) xi) De-pressurize to 145- 150 ksc and hold the pressure for removal of gags. Responsibility - HOD (O) xii) Remove the gags from all the spring-loaded safety valves of drum and superheater (all 08 no.) and communicate the same to UCB in writing. Responsibility - HOS (PP) xiii) During hydraulic test, visual inspection from outside boiler through peep hole and manhole may be done, but persons should enter the boiler only after Drum pressure is below 100 Ksc. Responsibility - HOS (PP) xiv) Units having history of weld and attachment failures should be tested by cyclic hydro. During cyclic hydro, system should be held at maximum allowable working pressure (of drum) for 30 min., then to be lowered by 100 ksc and held for 10 min. Again, pressure should be raised to maximum allowable working pressure and this cycle to be repeated three times. All activity to be recorded in pressure vs. time graph as per attached graph. Responsibility - HOD (O) & HOS (PP) xv) Reheater hydraulic leak test to be done after putting isolators , if reheater modification /major work in reheater is done. Responsibility - HOD (O) & HOS (PP)		
<u>8.0 EXCEPTION REPORTING</u>		
The exceptions encountered during hydraulic leak testing, against the above norms are to be reported as per the following format		
Responsibility - HOD (O) & HOS (PP)		
Unit No.	Description of exception	Remarks

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<u>9.0 RESPONSIBILITIES</u> The responsibility of carrying out different tasks in relation to precautions to be taken during hydraulic leak testing as detailed in the LMI are with All Shift Charge Engineers (Operation) and HOD (BMD). <u>10. RECORD KEEPING</u> The records to be maintained under this LMI include - Compliance record of norms detailed under this LMI by Operation - Exception reporting by BMD - Pressure parts - All Hydraulic leak test records and findings of hydro test inspection/rectification by BMD - Pressure Parts. <u>11.0 REVIEW</u> This document shall be reviewed by HOP (SIMHADRI) once in TWO years.		

NTPC		LMI		SIMHADRI
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<u>12.0 REVISION INDEX</u> (for changes/revision)				
Sl. No.	Rev. no.	Date	Details	Approved by
01	1.00	12.04.10	i) Approval routing changed at "LMI Approval document"	
			ii) Responsibility changed from SS (PP) to DGM (PP) at responsibilities	
			iii) Reviewing authority changed from DGM (MM) to GM I/c	
			iv) Distribution List	
02	2.00	22.04.13	i) Methodology of implementation is corrected in line with OGN	
			ii) Inclusion of U#3 & 4 in compliance record	
			iii) Responsibility changed from DGM (PP) to HOS (Pressure Parts) at Responsibilities	
			iv) Reviewing authority changed from GM I/c to HOP	
			v) Distribution List	
03	3.00	10.01.2020	i) Approval routing changed at "LMI Approval document"	
			ii) Review Frequency changed from "three years" to "TWO years"	
			iii) Distribution List	
04	4.00	20.03.2020	i) 02 Nos of LMIs: LMI/OGN/OPS/MECH/005 and LMI/OIN /OPS/MECH/001 documents combinedbased on Revised OGN - COS-ISO-00-OGN/OPS/MECH/005 Rev.No.1 ii) Hydraulic test water temperature as perIBR detailed in clause 7 (iii) iii) Provision for hydraulic test by reciprocating pump added at clause 7 (viii) iv) Drum Pressure restriction & entry of persons for inspection during hydraulic testspecified at clause 7(xiii) v) Cyclic hydro test procedure for unitswith history of weld and attachment failures detailed at clause 7(xiv)	
05	5.00	27.07.2020	i) Methodology implementation at Sl. No. 7 vii) Page 3 is corrected as Pressure raiseshould not be more than 10 ksc/min up to 80% of max. working pressure (of drum) and after that it should be limited to 2 ksc/min in line as per OGN ii) Distribution List	
06	5.00	05.07.2022	No changes done	
07	6.00	18.11.2022	Changes done as per COS-ISO-00-OGN/OPS/MECH/005,Rev. No.: 1 Oct. 2022	

[illegible]

NTPC	14.0 COMPLIANCE RECORD	SIMHADRI
TITLE: BOILER HYDRAULIC LEAK TEST DURING BREAKDOWN REPAIRS REV: 6.00 DATE: Nov'2022		
REF NO. LMI/OD/OPS/MECH/005/		PAGE : 8 OF 8
UNIT NO.	ACTIVITY	COMPLIANCE
I/II/III/IV		

